5. The trunk of a tree has diameter 30 cm. During the following year, the circumference grew 6 cm.
   (i) Approximately, how much did the tree's diameter grow?
   (ii) What is the percentage increase in area of the tree's cross-section?
6. An egg of a particular bird is very nearly spherical. If the radius to the inside of the shell is 5 mm and radius to the outside of the shell is 5.3 mm, find the volume of the shell approximately.
7. Assume that the cross section of the artery of human is circular. A drug is given to a patient to dilate his arteries. If the radius of an artery is increased from 2 mm to 2.1 mm, how much is cross-sectional area increased approximately?
8. In a newly developed city, it is estimated that the voting population (in thousands) will increase according to $V(t) = 30 + 12t^2 - t^3,\ 0 \le t \le 8$ where t is the time in years. Find the approximate change in voters for the time change from 4 to $4\frac{1}{6}$ year.
9. The relation between the number of words y a person learns in x hours is given by $y = 52\sqrt{x},\ 0 \le x \le 9$. What is the approximate number of words learned when x changes from
   (i) 1 to 1.1 hour? (ii) 4 to 4.1 hour?
10. A circular plate expands uniformly under the influence of heat. If it's radius increases from 10.5 cm to 10.75 cm, then find an approximate change in the area and the approximate percentage change in the area.
11. A coat of paint of thickness 0.2 cm is applied to the faces of a cube whose edge is 10 cm. Use the differentials to find approximately how many cubic centimeters of paint is used to paint this cube. Also calculate the exact amount of paint used to paint this cube.

8.3 Functions of Several Variables

Recall that given a function f of x; we sketch the graph of $y = f(x)$ to understand it better. Generally, the graph of $y = f(x)$ gives a curve in the xy-plane. Also, the derivative $f'(a)$ of f at $x = a$ represents the slope of the tangent at $x = a$, to the graph of f. In the introduction we have seen the need for considering functions of more than one variable. Here we shall develop some concepts to understand functions of more than one variable. First we shall consider functions of two variables. Let $F(x, y)$ be a function of x and y. To obtain graph F, we graph $z = F(x, y)$ in the xyz-space. Also, we shall develop the concepts of continuity, partial derivatives of a function of two variables.

Let us look at an example, $g(x, y) = 30 - x^2 - y^2$, for $x, y \in \mathbb{R}$. Given a point $(x, y) \in \mathbb{R}^2$, then $z = 30 - x^2 - y^2$ gives the z coordinate of the point on the graph. Thus the point $(x, y, 30 - x^2 - y^2)$ lies $30 - x^2 - y^2$ high just above the point (x, y) in xy-plane. For instance, for $(2,3) \in \mathbb{R}^2$, the point $(2,3,30 - 2^2 - 3^2) = (2,3,17)$ lies on the graph of g. If we fix the value $y = 3$, then $g(x,3) = -x^2 + 21$ which is a function that depends only on x variable; so its graph must be a curve. Similarly, if we fix value $x = 2$, then we have $g(2, y) = 26 - y^2$ which is a function that depends only on y. In each case the graph, as the resulting function being quadratic, will be a parabola. The surface we obtain from $z = g(x, y)$ is called paraboloid.

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